

Artificial Intelligence

Haystack Dense Retrieval Practice Document

Introduction

Artificial Intelligence (AI) is the field of computer science concerned with creating systems that can perform tasks that normally require human intelligence. These tasks include reasoning, learning, perception, language understanding, planning, and decision making.

Major Areas

Important areas of AI include machine learning, natural language processing, computer vision, robotics, expert systems, speech processing, and planning. Generative AI is a modern area that can create text, images, audio, code, and other content.

AI and Machine Learning

Machine learning is a major approach used to build AI systems. AI is the broader concept of intelligent computer behavior, while ML focuses on learning useful patterns from data. Deep learning is a subset of ML based on multi-layer neural networks.

Applications

AI is used in virtual assistants, search engines, autonomous systems, recommendation engines, healthcare support, financial risk analysis, cybersecurity, education, and business automation.

Challenges

AI systems can face challenges involving bias, privacy, security, reliability, explainability, data quality, and responsible use. Human oversight and appropriate evaluation are important when AI is used for high-impact decisions.

This document is prepared as a small knowledge source for a Haystack BM25 versus dense retrieval experiment.